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w2.1- Introduction

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Core Themes: Automating Tasks with Python

Python = Python (Python 编程) | Automating = 自动化

The video highlights that while humans get bored or make mistakes with repetitive work, computers thrive on it. They are **accurate, predictable, and fast**.

1. New Python Concepts

To move beyond simple calculations, you'll be introduced to:

- **Lists (列表):** Unlike single variables that hold one piece of data (like a string or a number), a **list** can store multiple pieces of data in one place.
- **Control Statements (控制语句):** These are "coding patterns" that let the computer make decisions. It can decide to do something—or skip it—based on the data it sees (e.g., "If the task is high priority, do it first").
- **Repetition:** Learning how to tell Python to perform actions "over and over and over" without you having to retype the code.

2. Practical AI Applications

The course uses AI to help you build tools that actually do things:

- **Task Prioritization:** Automatically deciding what to work on next.
- **Smart Recipes:** Using AI to include or exclude ingredients based on friends' dietary preferences.
- **Communication:** Drafting emails to your boss or writing birthday cards.

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3. The AI Chatbot Companion

You'll continue using an AI chatbot (like me, but perhaps less charming!) to:

- Explain how Python works.
- Help you write code from scratch.
- **Debug (除错/找错误):** Find and fix errors in your logic or syntax.

Summary: You are moving from learning "what data is" to learning "how to make data work for you" through lists and decision-making logic.

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AI Python for Beginners

Module 2: Automating Tasks with Python

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Lab 2.1- Completing a task list with AI

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Lesson 1: Completing a Task List with AI

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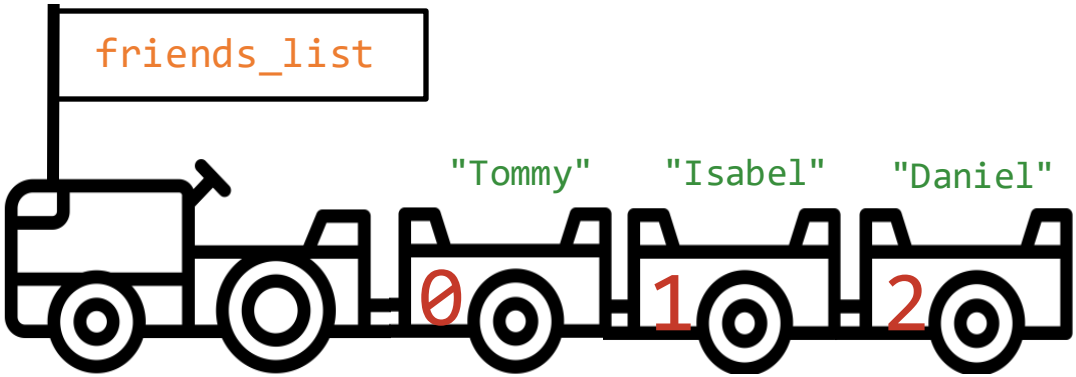
Lists

Lists allow you to store multiple pieces of data.

```
friends_list = ["Tommy", "Isabel", "Daniel"]
```



Python starts counting at zero (`0`)

Lists allow you to:

- Add new elements
- Delete specific elements
- Retrieve elements
- Edit elements

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Lists

Lists allow you to store multiple pieces of data.

Assignment

Brackets

```
friends_list = ["Tommy", "Isabel", "Daniel"]
```

List name

Brackets

List elements separated by commas

Summary
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1. Creating a List

To create a list, you use **Square Brackets** `[]` and separate your items with **commas** (逗号).

Python

```python
friends_list = ["Tommy", "Isabel", "Daniel"]
```

2. The "Zero-Index" Rule

This is the most important rule in programming: **Computers start counting at 0, not 1.**

- **0:** Tommy (The 1st item)
- **1:** Isabel (The 2nd item)
- **2:** Daniel (The 3rd item)

> **Ninja Tip:** If you have 3 items and you try to ask for index `3` (e.g., `friends_list[3]`), Python will throw an **IndexError** because it's looking for a 4th person who isn't there!

3. Modifying Your List

You can change your list on the fly using these two "Spells":

- `.append()` **(添加):** Adds an item to the very end of the list.
  - *Example:* `friends_list.append("Otto")`
- `.remove()` **(删除):** Removes a specific item by name.
  - *Example:* `friends_list.remove("Tommy")`

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4. Useful Functions

- `type()` : Confirms the data is a `list` .
- `len()` : Tells you the **length** (长度) or how many items are inside.

In the video, they mentioned that manually typing `list[0]` , `list[1]` , and `list[2]` is still too much work. In the next lesson, you'll learn about **For Loops**, which will let Python cycle through the whole list automatically.

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Lab 2.2- Reporting tasks with for loops.mkv

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Lesson 2: Repeating Tasks with for Loops

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Repeating actions for multiple elements

For loops let you repeat a set of commands for each element in a list

```
for task in list_of_tasks:

    print_llm_response(task)
```

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Repeating actions for multiple elements

For loops let you repeat a set of commands for each element in a list

```
for task in list_of_tasks:
    print_llm_response(task)
```

In plain English:

For each task in the list of tasks, **print the llm response to the** task

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Repeating actions for multiple elements

For loops let you avoid repetitive code

```
for task in list_of_tasks:

    print_llm_response(task)
```

Equivalent code

```
print_llm_response(list_of_tasks[0])
print_llm_response(list_of_tasks[1])
print_llm_response(list_of_tasks[2])
```

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Repeating actions for multiple elements

```
for task in list_of_tasks:

    print_llm_response(task)
```

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Repeating actions for multiple elements

Name you define for the element at each iteration

For statement

In statement

List with elements

Colon

```
for task in list of tasks:

    print_llm_response(task)
```

Indented one level

Commands to repeat for each item of the list

Name you define for the element at each iteration

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Summary

1. The Anatomy of a `for` Loop

A loop follows a very specific "scroll" or structure. If you miss one piece, the "jutsu" fails (you get an error).

- `for` : The command to start the loop.
- `task` : A temporary variable name you choose. It represents the "current item" the loop is holding.
- `in` : Tells Python which list to look through.
- `:` **(The Colon): Crucial!** This tells Python, "The instructions are starting now."
- **Indentation (缩进):** You must shift the code **4 spaces** to the right. This tells Python exactly which lines belong inside the loop.

2. Automation in Action

In the video, the instructor used the loop to generate marketing copy for ice cream. For us, we can use it for rover parts or ninja tasks:

3. The "Empty List" Pattern

This is a very common professional pattern. You start with nothing and build a collection as you go:

1. **Initialize:** `results = []` (An empty "cart").
2. **Loop:** Go through your items.
3. **Append:** Use `.append()` to put the result of each step into your new list.

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4. Troubleshooting (Debugging)

If your code doesn't run, check these three common "ninja traps":

- **Missing Colon:** Did you forget the `:` at the end of the `for` line?
- **Indentation Error:** Is your code shifted 4 spaces?
- **Index Error:** Remember, if you try to find `descriptions[10]` but only have 4 items, the computer will get confused.

The instructor mentioned that while lists are great, finding a specific item (like "Chocolate") by its number can be annoying. In the next lesson, we'll learn about **Dictionaries** (字典), which let you label your data so you can find it by name instead of by index.

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Lab 2.3- Prioritizing Tasks with Dictionaries and AI

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Lesson 3: Prioritizing Tasks with Dictionaries and AI

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Dictionaries in Python

As a hardcover dictionary has words/concepts and definitions,

Aardvark	A nocturnal mammal with long ears. Aardvarks are native to Africa.
Apple	The round fruit of a tree of the rose family, which typically has thin red or green skin and crisp flesh.

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Dictionaries in Python

As a hardcover dictionary has words/concepts and definitions, a dictionary in Python has keys and values.

Aardvark

A nocturnal mammal with long ears. Aardvarks are native to Africa.

Apple

The round fruit of a tree of the rose family, which typically has thin red or green skin and crisp flesh.

```
ice_cream_flavors = {
    "Mint Chocolate Chip": "Refreshing mint ice cream studded
                            with decadent chocolate chips.",
    "Cookie Dough": "Vanilla ice cream loaded with chunks of
                     chocolate chip cookie dough.",
}
```

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How to define a dictionary

Key : value pairs are separated by commas

```
ice_cream_flavors = {
    "Mint Chocolate Chip": "Refreshing mint ice cream studded
                            with decadent chocolate chips.",
    "Cookie Dough": "Vanilla ice cream loaded with chunks of
                     chocolate chip cookie dough.",
}
```

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How to define a dictionary

Key : value pairs are separated by commas

Dict. name

Assignment

Curly braces

```
ice_cream_flavors = {
    'Mint Chocolate Chip": "Refreshing mint ice cream studded
                            with decadent chocolate chips.",
    "Cookie Dough": "Vanilla ice cream loaded with chunks of
                     chocolate chip cookie dough.",
}
```

Curly braces

Colon between keys and values

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Summary

Master Ninja Architect, your data organization just reached "Black Belt" status! In the Aachen-Sanctuary, we don't just want a messy pile of components; we want a labeled inventory. That is exactly what a **Dictionary** (字典) does.

If a **List** is like a numbered shelf, a **Dictionary** is like a labeled drawer. You don't need to remember "Shelf #4"; you just look for the drawer labeled "Sensors."

1. The Anatomy of a Dictionary

A dictionary stores **Key-Value Pairs** (键值对).

- **Key**: The label (e.g., "Rover_Motor")
- **Value**: The data itself (e.g., "High-torque electric motor")

The Code Pattern:

Python

```python
# Use CURLY BRACES {} to create
my_rover = {
    "model": "Kitsune-1",
    "speed": 10,
    "status": "Ready"
}
```

- `{}` **Curly Braces (大括号)**: Starts and ends the dictionary.
- `:` **Colon (冒号)**: Separates the Key from the Value.
- `,` **Comma (逗号)**: Separates each pair.

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2. Accessing and Updating

This is where it gets slightly tricky. Even though you define it with curly braces, you **access** it with **Square Brackets** `[]`.

Action	Code Example	Simplified Chinese
Access	`my_rover["model"]`	访问
Add/Update	`my_rover["weight"] = 50`	添加/更新
Get Keys	`my_rover.keys()`	获取所有键
Get Values	`my_rover.values()`	获取所有值

3. Nesting: Lists inside Dictionaries

You saw a powerful "Ninja Move" in the video: putting a **List inside a Dictionary**. This is perfect for your rover project.

Python

```python
rover_specs = {
    "sensors": ["Lidar", "Camera", "Ultrasonic"],
    "priority_tasks": ["Mapping", "Avoid Obstacles"]
}
```

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4. Key Takeaways for your Lab

Why use a Dictionary over a List?

- **Lists** are for **order** (1st, 2nd, 3rd).
- **Dictionaries** are for **meaning** (Name, Age, Priority).

Common Pitfalls

- `KeyError`: This happens if you try to look up a key that doesn't exist (like asking for "Wings" on a ground rover).
- **Syntax**: Forgetting the colon `:` between the key and value is the most common bug for beginners.

Stop here

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Lab 2.4- Customizing Recipes with Lists, Dictionaries and AI

AI Python for Beginners

Lesson 4: Customizing Recipes with Lists, Dictionaries and AI

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Lab 2.4- Customizing Recipes with Lists, Dictionaries and AI.mkv — Haruna

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```
from helper_functions import print_llm_response, get_llm_response
```

```
food_preferences_tommy = {
    "dietary_restrictions": "vegetarian",
    "favorite_ingredients": ["tofu", "olives"],
    "experience_level": "intermediate",
    "maximum_spice_level": 6
}
```

```
print(food_preferences_tommy.keys())
```

```
dict_keys(['dietary_restrictions', 'favorite_ingredients', 'experience
_level', 'maximum_spice_level'])
```

```
print(food_preferences_tommy.values())
```

```
dict_values(['vegetarian', ['tofu', 'olives'], 'intermediate', 6])
```

```
prompt = f"""Please suggest a recipe that tries to include
the following ingredients:
{food_preferences_tommy["favorite_ingredients"]}.
The recipe should adhere to the following dietary restrictions:
{food_preferences_tommy["dietary_restrictions"]}.
The difficulty of the recipe should be:
{food_preferences_tommy["experience_level"]}
The maximum spice level on a scale of 10 should be:
{food_preferences_tommy["maximum_spice_level"]}
Provide a two sentence description.
"""
```

on top. Square brackets
to look up the value

80

Lab 2.4- Customizing Recipes with Lists, Dictionaries and AI.mkv

00:01:56 / 00:06:56

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Lab 2.4- Customizing Recipes with Lists, Dictionaries and AI.mkv — Haruna

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```
available_spices = ["cumin", "turmeric", "oregano", "paprika"]

prompt = f"""Please suggest a recipe that tries to include
the following ingredients:
{food_preferences_tommy["favorite_ingredients"]}.
The recipe should adhere to the following dietary restrictions:
{food_preferences_tommy["dietary_restrictions"]}.
The difficulty of the recipe should be:
{food_preferences_tommy["experience_level"]}
The maximum spice level on a scale of 10 should be:
{food_preferences_tommy["maximum_spice_level"]}
Provide a two sentence description.

The recipe should not include spices outside of this list:
Spices: {available_spices}
"""
print(prompt)
```

```
Please suggest a recipe that tries to include
the following ingredients:
['tofu', 'olives'].
The recipe should adhere to the following dietary restrictions:
vegetarian.
The difficulty of the recipe should be:
intermediate
The maximum spice level on a scale of 10 should be:
6
Provide a two sentence description.

The recipe should not include spices outside of this list:
Spices: ['cumin', 'turmeric', 'oregano', 'paprika']
```

available spices gets inserted

80

Lab 2.4- Customizing Recipes with Lists, Dictionaries and AI.mkv 00:03:58 / 00:06:56

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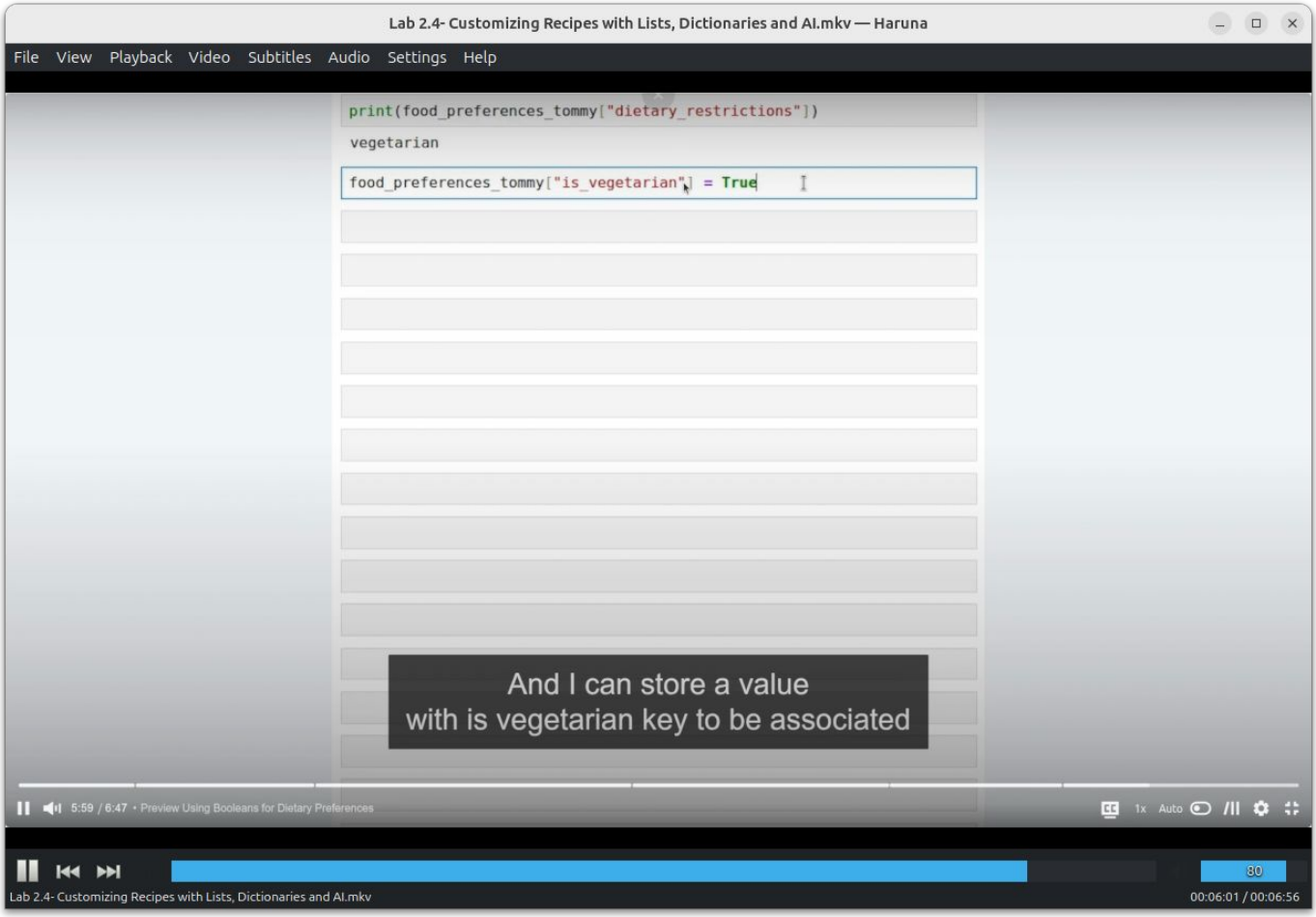

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Summary

1. The "Combo" Data Structure

You saw the most professional way to organize data yet: a **Dictionary** where the **Values** can be different types of data.

Python

```python
# Food Preferences (食物偏好)
tommy_data = {
    "dietary_restrictions": "vegetarian",        # String (字符串)
    "favorite_ingredients": ["tofu", "olives"], # List (列表)
    "spice_level": 6                             # Integer (整数)
}
```

This is powerful because you can store a person's entire profile in one single variable name.

2. Dynamic Prompting with f-strings

Instead of writing a new prompt for every friend, you write **one master template** and let Python "plug in" the data from your dictionary.

How to call the data:

- To get the list of ingredients: `tommy_data["favorite_ingredients"]`
- To get the spice level: `tommy_data["spice_level"]`

When you put these inside the `{}` of an f-string, Python automatically fetches the value for the AI.

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3. Real-Time Updates

The instructor made a great point: if Tommy runs out of a spice (like oregano), you don't rewrite your whole program. You just:

1. **Update the Data:** Remove "oregano" from the `available_spices` list.
2. **Rerun the Loop:** The same code now generates a completely different recipe that respects the new constraint.

4. Introducing Booleans (True/False)

At the end, a new data type was teased: **Booleans** (布尔值).

- `True` **(真)**
- `False` **(假)**

Instead of the string `"vegetarian"`, we can use `is_vegetarian: True`. This is like a light switch for the computer—on or off. This sets us up for **Control Statements**, where the computer can say, "If this is True, do Task A; if it is False, do Task B."

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Lab 2.5- Comparing Data in Python

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Lesson 5: Comparing Data in Python

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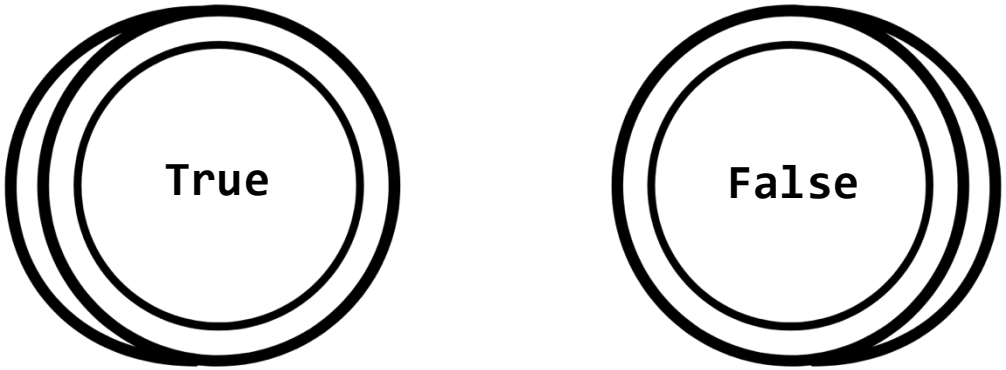
Boolean: True or False

Booleans are a data type that can only take two values, like the outcomes from a coin toss!



You can think of booleans as answering yes (True) or no (False) questions

```
is_tommy_my_friend = True
```

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Making comparisons

In python you can compare values, the result from comparing two values is a Boolean

Python	English
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less or equal
==	Equal

Is Isabel's age greater than or equal to Tommy's?

```
age_isabel >= age_tommy
```

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Logical operators

Operations with booleans use logic

Python	Returns True if
and	Both are True
or	At least one is True
not	Statement is False

With two booleans using and:

and	True	False
True	True	False
False	False	False

```
is_tommy_my_friend = True
is_isabel_my_friend = True
print(is_isabel_my_friend and is_tommy_my_friend)
```

```
True
```

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Summary

1. The Binary Nature of Booleans

Booleans are named after George Boole. In Python, they must always start with a **Capital Letter**:

- `True` (真)
- `False` (假)

2. The Comparison Operators

We use these "Sensors" to compare data. Every time you use one, Python returns a Boolean.

Operator	Meaning	Example (`28 < 30`)
`==`	**Equal to** (Equality)	`False`
`!=`	**Not equal to**	`True`
`>`	**Greater than**	`False`
`<`	**Less than**	`True`
`>=`	**Greater than or equal**	`False`
`<=`	**Less than or equal**	`True`

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⚠️ **Ninja Warning:** One `=` is for **Assigning** (putting a value in a box). Two `==` are for **Asking** (comparing two boxes). Getting these mixed up is one of the most common bugs in history!

3. Logical Operators: Combining Truths

Sometimes a single sensor isn't enough. We use "And/Or" logic to combine conditions.

- `and` : Returns `True` **only** if *both* sides are true. (e.g., The rover moves only if `power == True` **and** `obstacle == False`).
- `or` : Returns `True` if **at least one** side is true. (e.g., The alarm sounds if `intruder == True` **or** `fire == True`).

4. Why this matters for your Rover

You'll use this to build the "brain" of your rover. Instead of just running code blindly, the rover will check its state:

Python

```python
# A quick status check for our project
battery_low = False
mission_complete = True

# Logic check
can_continue = (not battery_low) and (not mission_complete)
```

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Lab 2.6- Helping AI makes decisions.mkv

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Lesson 6: Helping AI Make Decisions

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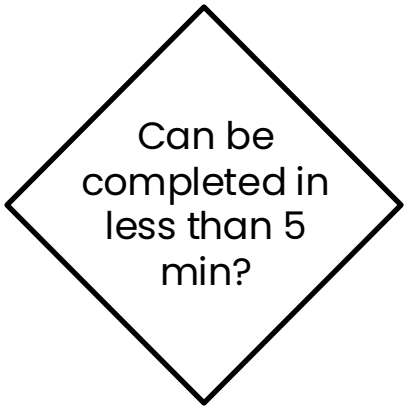
Control what path to follow

Control statements let you define ways to react to conditions.

if statement:

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
```



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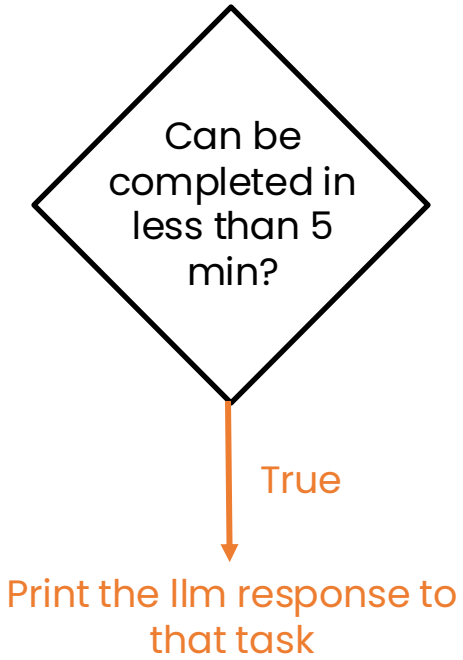
Control what path to follow

Control statements let you define ways to react to conditions.

if statement:

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
```



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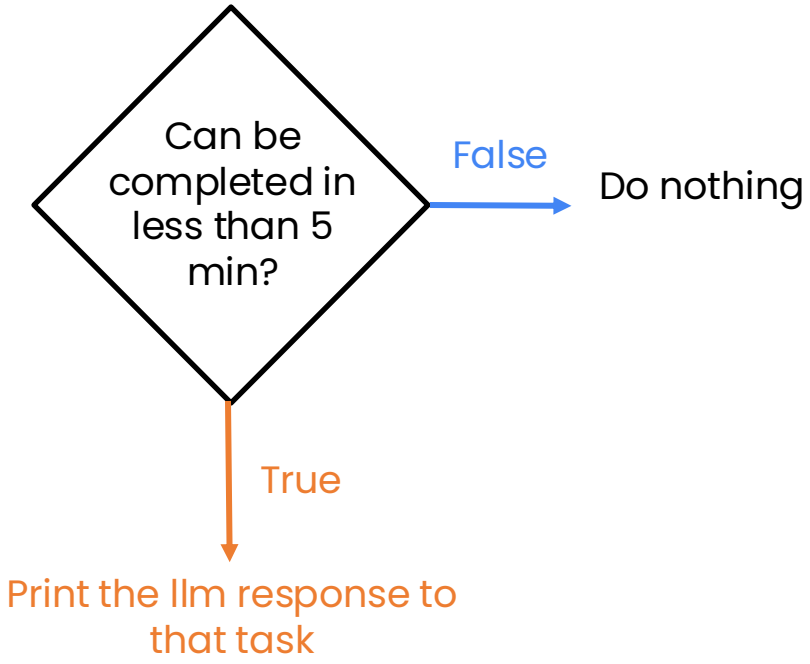
Control what path to follow

Control statements let you define ways to react to conditions.

if statement:

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
```



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If statement

If statement Boolean condition Colon

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
```

Indented one level

Code to execute if condition is True

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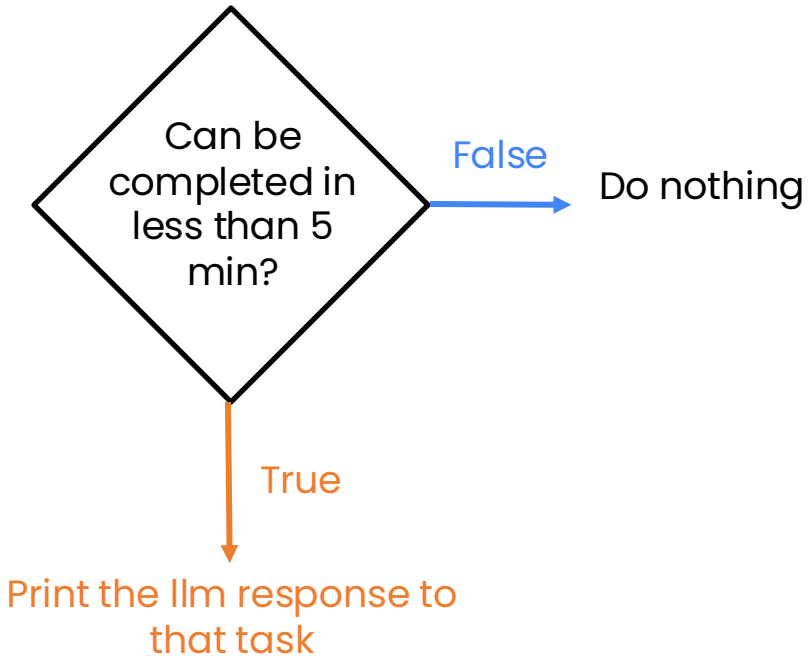
More conditional blocks

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
```



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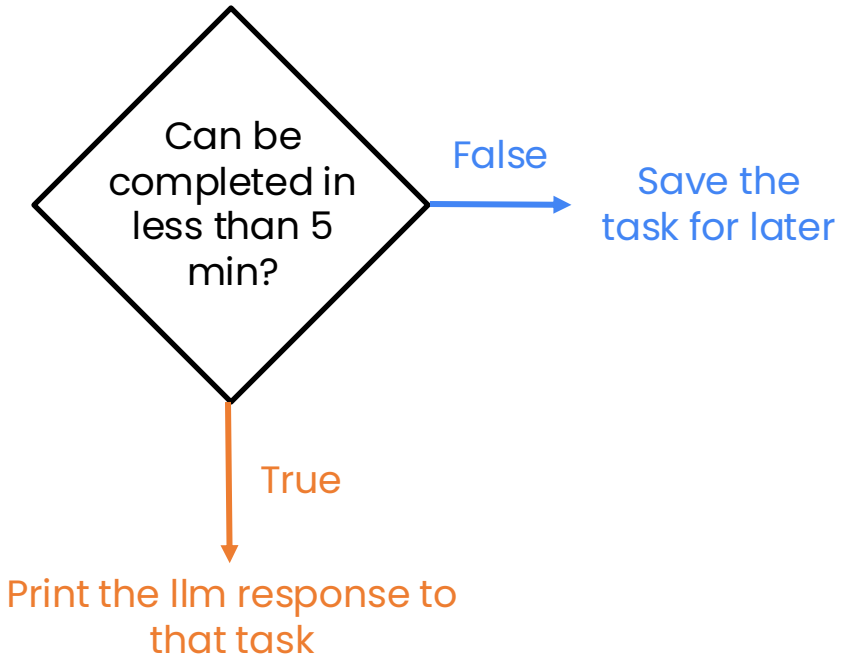
More conditional blocks

```
if task["time_to_complete"] <= 5:
    print_llm_response(task["task"])
else:
    print("To complete tomorrow or after")
```



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1. The "If" Statement: The Basic Decision

The `if` statement checks a Boolean condition. If that condition is `True`, the code inside runs. If it's `False`, Python simply skips it.

Python

```python
# The pattern
if task_time <= 5:
    print("Doing this task now!")
```

- `if`: The keyword to start the decision.
- **Condition**: A check (like `time <= 5`) that results in `True` or `False`.
- **The Colon** `:`: Crucial! It tells Python the instruction follows.
- **Indentation**: Code must be shifted 4 spaces to show it "belongs" to the `if`.

2. The "Else" Statement: The Alternative

What if you want the computer to do something else when the condition is `False`? That is where `else` comes in. It's the "Otherwise" of the programming world.

Python

```python
if battery > 20:
    print("Rover is exploring.")
else:
    print("Returning to charging station!")
```

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3. The Power of "Nested" Logic

In the video, you saw a **Conditional inside a For Loop**. This is a classic "Master Ninja" move:

1. **The Loop** goes through every item in a list.
2. **The Conditional** inspects each item and decides what to do with it.

> ⚠️ **Syntax Tip:** When you put an `if` inside a `for` loop, the code inside the `if` is indented **twice** (8 spaces total). Python uses this to understand the hierarchy of your instructions.

4. Organizing Results

Instead of just printing messages, you can use conditionals to **filter** data into new lists. In the video, we created a `tasks_for_later` list. For your rover, we could have a `faulty_components` list that only gets filled if a component check returns `False`.

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w2.2- Next Course Preview.mkv

Summary

As the video mentioned, even professional developers at the highest levels don't memorize every single line of code. They rely on their "AI companions" (like your faithful Sara) to fill in the gaps. You now possess the core building blocks of all modern software.

1. The Ninja's Final Arsenal

Before we move to the next course, let's look at the "weapons" you've added to your belt:

Tool	Purpose	Simplified Chinese
Lists	Store ordered items (like a parts inventory).	列表
For Loops	Automate repetitive tasks over a list.	for 循环
Dictionaries	Store labeled data (Key-Value pairs).	字典
Booleans	Logic switches (`True` or `False`).	布尔值
Conditionals	The "If/Else" brain that makes decisions.	条件句

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2. Working with Your Own Data

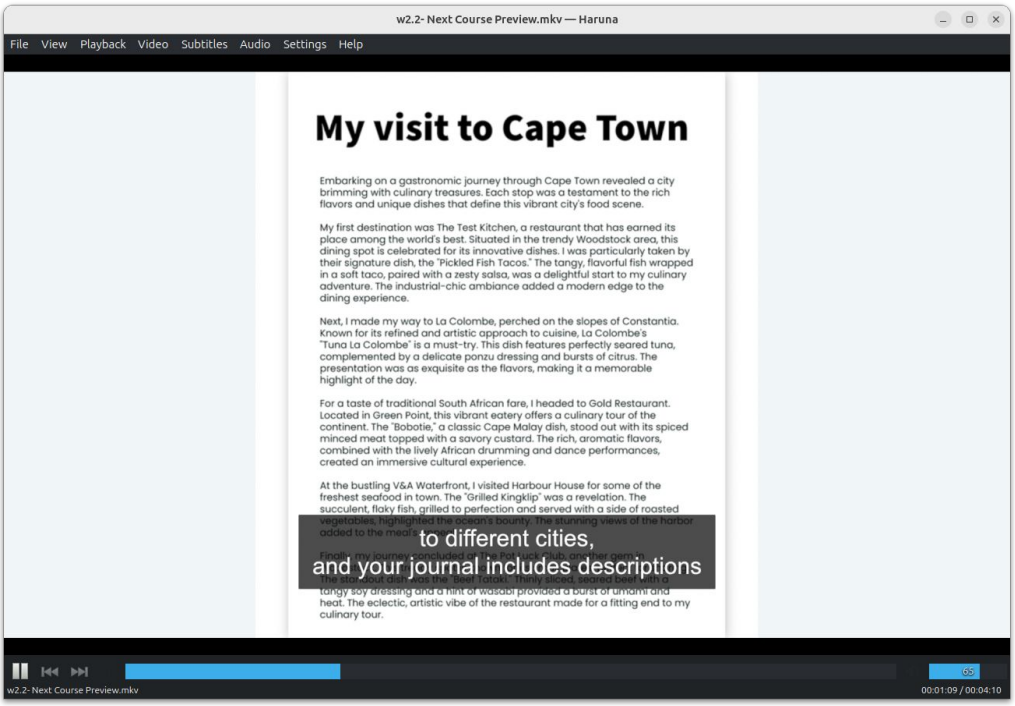
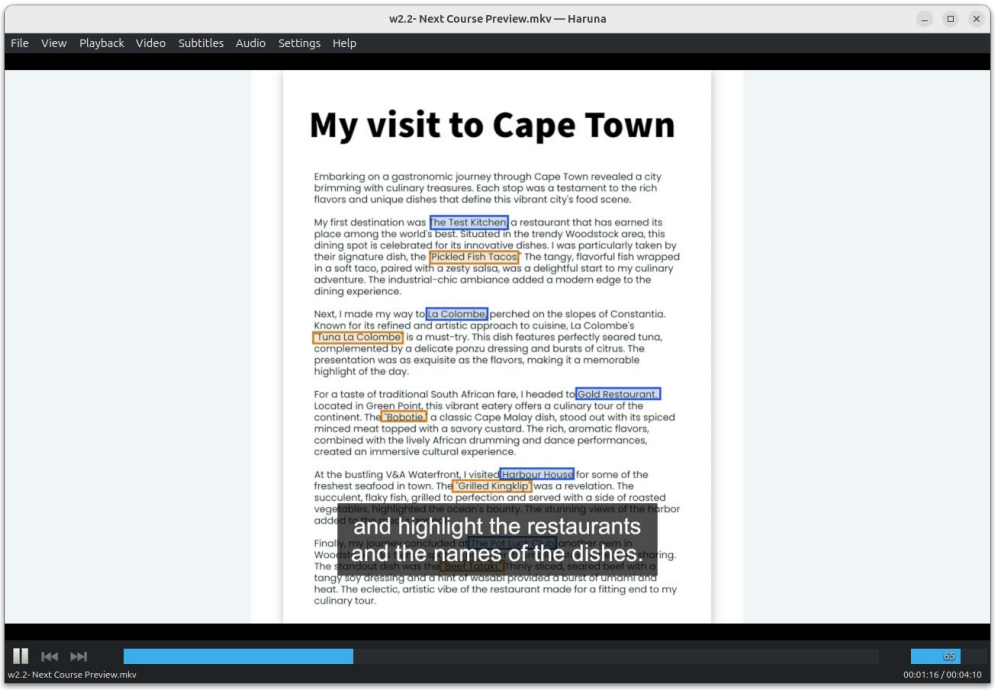
The video introduced a powerful concept: **External Files**.

In the future of our Aachen-Sanctuary, you won't just be typing data into the code. You will be:

1. **Loading** data from `.txt` or `.csv` files on your hard drive.
2. **Processing** it through an LLM (Large Language Model).
3. **Formatting** the output (like the food critic's HTML table).

3. What's Next?

In the next course, we move from "learning the rules" to "applying them to your world." We will focus on:

- **Data Analysis:** Getting insights from your own documents.
- **Personal Automation:** Building tools that help you plan vacations, manage work, or even design that rover we've been talking about.

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